SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 2 – Case Study

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO3 – Precision	**Assessment:**	Assessment Tool 2 – Case Study
Weightage:	40% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
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Section / Batch:	AI & ML / 2024	**Date:**	06/08/2026

Code of Conduct

I _D. Sri Anjaneya / 192425219_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _D. Sri Anjaneya_

Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

Section / Topic	Focus	Case Study
N-Gram Language Models and Smoothing Techniques	Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction	Case Study 1 – Smart Mobile Keyboard Prediction System
Part-of-Speech (POS) Tagging and Word Classes	Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding	Case Study 2 – AI-Powered Customer Support Chatbot
Transformation-Based Tagging (Brill Tagger)	POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling	Case Study 3 – News Analytics and POS Tag Correction System

Corpus Statistics and Probabilistic Analysis	Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation	Case Study 3 – News Analytics and POS Tag Correction System
Integrated Syntactic Analysis and NLP Applications	Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems	Case Studies 1, 2, and 3

Annexure A – Case Study

CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus: *"Data science is powerful, data science drives innovation, data science is evolving."*
The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences. The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- λ = 0.5 (Trigram)
- λ_{123} = 0.3 (Bigram)
- λ = 0.2 (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.

3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.
Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging. Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.
The sentence processed by the system is:
"Economic growth increases employment." Initial
POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

Change NNS to VBZ if the preceding word is tagged as NN.

The system also performs corpus-based frequency analysis and uncertainty evaluation.

Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
   - economic (120 occurrences)
   - growth (450 occurrences)
   - increases (210 occurrences)
   - employment (380 occurrences)

   Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

CASE STUDY 1: Smart Mobile Keyboard Prediction System

1. Bigram Probability Calculation using MLE:

Given:

- $C(\text{data}) = 3$
- $C(\text{data science}) = 3$

Formula:

$$P(\text{science} \mid \text{data}) = \frac{C(\text{data science})}{C(\text{data})}$$

Calculation:

$$P(\text{science} \mid \text{data}) = \frac{3}{3} = 1.0$$

Answer:

$$\boxed{P(\text{science} \mid \text{data}) = 1.0}$$

Interpretation:

The probability **1.0** shows that every occurrence of the word **"data"** is followed by **"science"** in the given corpus. This indicates a strong relationship between the two words.

In a smart mobile keyboard, when a user types **"data"**, the system predicts **"science"** with maximum confidence. This improves typing speed, reduces keystrokes, and provides accurate next-word suggestions.

2. Backoff Model for Unseen Word Sequence:

Given Sequence:

"data science improves"

The trigram sequence is not available in the training corpus.

Calculation:

First, the model checks trigram probability:

$$P(i, mproves \mid datascience) = 0$$

Since the trigram is unavailable, it backs off to bigram:

$$P(improves \mid science) = 0$$

The model then checks unigram probability:

$$P(improves) = 0$$

Answer:

The estimated probability is very low because the word **"improves"** is not present in the given corpus.

Analysis:

The Backoff Model handles unseen word combinations by using lower-order n-grams. It prevents the language model from completely failing when users enter new sentences.

Application:

In predictive keyboards, backoff helps generate suggestions even for unknown word sequences and improves real-time text prediction.

3. Deleted Interpolation Probability Calculation:

Given Weights:

- Trigram weight = 0.5
- Bigram weight = 0.3
- Unigram weight = 0.2

For the sequence:

$$P(i, s \mid datascience)$$

Trigram Probability:

$$P(i, s \mid datascience) = \frac{C(data\ science\ is)}{C(data\ science)}$$

$$= \frac{2}{3} = 0.66$$

Bigram Probability:

$$P(is \mid science) = \frac{2}{3} = 0.66$$

Unigram Probability:

$$P(is) = \frac{2}{12} = 0.16$$

Interpolation Calculation:

$$P = 0.5(0.66) + 0.3(0.66) + 0.2(0.16)$$
$$P = 0.33 + 0.198 + 0.032$$
$$P = 0.56$$

Answer:

$$\boxed{P(i, s \mid datascience) = 0.56}$$

Analysis:

Deleted Interpolation combines different n-gram probabilities to improve prediction accuracy. It balances higher-order information with lower-order word frequency.

Application:

It helps keyboards make reliable predictions even when the available training data is limited.

4. Entropy Analysis of Word Prediction:

Given:

$$P(is) = 0.66$$
$$P(drives) = 0.33$$

Formula:

$$H(X) = -\sum P\,(x)\log_2 P(x)$$

Calculation:

$$H(X) = -\big(0.66\log_2(0.66) + 0.33\log_2(0.33)\big)$$
$$H(X) = 0.923$$

Answer:

$$\boxed{Entropy = 0.923 \text{ bits}}$$

Interpretation:

The entropy value represents uncertainty in prediction. A lower entropy value means the model is more confident, while higher entropy indicates uncertainty.

Since **"is" (0.66)** has a higher probability than **"drives" (0.33)**, the keyboard has better confidence in predicting **"is"**.

Application:

Entropy helps smart keyboards measure prediction confidence and decide the most suitable word suggestions for users.

CASE STUDY 2: AI-Powered Customer Support Chatbot

1. POS Tagging of Sentences:

Sentence 1: "Book a flight ticket now."

Word	**POS Tag**	**Meaning**
Book	VB	Verb
a	DT	Determiner
flight	NN	Noun
ticket	NN	Noun
now	RB	Adverb

Sentence 2: "This book is interesting."

Word	**POS Tag**	**Meaning**
This	DT	Determiner
book	NN	Noun
is	VBZ	Verb
interesting	JJ	Adjective

Analysis:

The word **"book"** has different meanings depending on the context.

- In **"Book a flight ticket now"**, "book" represents an action (making a reservation), so it is tagged as **VB (Verb)**.
- In **"This book is interesting"**, "book" represents an object, so it is tagged as **NN (Noun)**.

Context helps the chatbot understand the correct grammatical role of ambiguous words.

2. HMM Probability Calculation for "Book" as Verb:

Given:

$$P(book \mid VB) = 0.6$$
$$P(Start \rightarrow VB) = 0.5$$

Formula:

$$P(VB \mid book) = P(Start \rightarrow VB) \times P(book \mid VB)$$

Calculation:

$$= 0.5 \times 0.6$$
$$= 0.3$$

Answer:

$$\boxed{P(Book\ as\ VB) = 0.3}$$

Analysis:

The probability of "book" being a verb is higher because the sentence begins with an action word followed by an object ("a flight ticket"). The HMM model identifies this pattern and selects the **VB tag**.

Application:

Probabilistic tagging helps chatbots correctly understand user intentions. In this case, identifying "book" as a verb helps the chatbot understand that the user wants to reserve a flight.

3. Comparison of Rule-Based and HMM Tagging:

Rule-Based Tagging	HMM Tagging
Uses manually created grammar rules.	Uses probability from training data.
Difficult to handle new sentences.	Handles unknown patterns better.
Requires human effort to create rules.	Learns patterns automatically.
Less flexible for large datasets.	More suitable for large-scale applications.

Analysis:

Rule-based systems work well for simple and fixed grammar structures, but they struggle with ambiguous words. HMM tagging uses statistical information and context, making it more effective for chatbot applications.

Recommendation:

HMM-based tagging is more suitable for large-scale chatbot deployment because it can process large amounts of text, learn from data, and resolve word ambiguity efficiently.

4. Role of POS Tagsets and Word Classes in Chatbots

Analysis:

Standardized POS tagsets like the **Penn Treebank Tagset** provide a common method to identify grammatical roles of words. They help chatbots understand sentence structure and user intent.

Applications:

- Helps identify important keywords in user messages.
- Improves intent detection and classification.
- Helps generate grammatically correct responses.

Conclusion:

POS tagging and word classification are important components of chatbot systems. They allow AI models to understand language better and provide more accurate and meaningful responses.

CASE STUDY 3: News Analytics and POS Tag Correction System

1. Transformation-Based POS Tag Correction:

Given Sentence:

Economic growth increases employment.

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

Transformation Rule:

Change **NNS to VBZ** if the preceding word is tagged as **NN**.

Applying Rule:

The word **"growth"** is tagged as **NN** and the next word **"increases"** is tagged as **NNS**.

Therefore:

$$increases/NNS \rightarrow increases/VBZ$$

Updated POS Sequence:

- economic/JJ
- growth/NN
- increases/VBZ
- employment/NN

Analysis:

The correction is linguistically appropriate because **"increases"** acts as a verb in the sentence. The sentence structure is:

Subject + Verb + Object

- Economic growth → Subject
- increases → Verb
- employment → Object

The Brill Tagger improves the initial tagging by applying learned correction rules.

2. Analysis of Initial Tagging Errors:

Initial Error:

The word **"increases"** was incorrectly tagged as **NNS (Plural Noun)**.

Correct Tag:

$$increases/VBZ$$

Reason:

The word "increases" describes an action performed by the subject **"economic growth."** Therefore, it functions as a third-person singular verb.

Correct Grammatical Structure:

Word	**Correct Tag**	**Role**
Economic	JJ	Adjective
growth	NN	Subject Noun
increases	VBZ	Verb
employment	NN	Object Noun

Conclusion:

Transformation-based tagging corrects mistakes by using grammatical patterns and contextual information.

3. Word Frequency Distribution Analysis:

Given Corpus Frequencies:

Word	Frequency
economic	120
growth	450
increases	210
employment	380

Total Frequency:

$$120 + 450 + 210 + 380 = 1160$$

Frequency Distribution:

$$P(word) = \frac{Frequency(word)}{Total\ Frequency}$$

Word	Probability
economic	$120/1160 = 0.103$
growth	$450/1160 = 0.388$
increases	$210/1160 = 0.181$
employment	$380/1160 = 0.328$

Analysis:

Word frequency helps POS tagging models estimate the likelihood of words appearing in different grammatical roles. Frequently occurring words provide stronger statistical evidence for prediction.

Application:

Corpus frequency information improves probabilistic tagging accuracy by helping the system select the most likely POS tag based on learned patterns.

4. Effectiveness of Transformation-Based Tagging and Entropy Analysis:

Analysis:

Transformation-Based Tagging improves POS accuracy by applying correction rules after initial tagging. Instead of manually tagging every word, the system learns common errors and automatically fixes them.

Advantages:

- Corrects incorrect initial POS tags.
- Uses context-based rules.
- Improves tagging accuracy.
- Reduces human effort.

Entropy Analysis:

Entropy measures uncertainty in POS tag selection.

- **High entropy:** Multiple tags have similar probabilities, creating uncertainty.
- **Low entropy:** One tag has a higher probability, creating confidence.

Before applying the transformation rule, the tag **NNS** creates uncertainty for the word **"increases."** After correction to **VBZ**, the uncertainty decreases because the grammatical role becomes clear.

Conclusion:

Brill Tagger improves POS tagging performance by correcting errors using learned transformations. Entropy helps evaluate the confidence of tag assignments before and after correction.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Q. No. / Criteria Level	Understanding of Case	Application of Concepts	Analysis	Solution Design	Clarity	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____